Scope of Work

Description

This document provides a concise outline of the *AI Recommendation Filter* Chrome extension project, detailing its objectives, value distribution, core features, technology stack, deliverables, timeframe, optional features, exclusions, and success criteria under the assumption of smooth development.

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1.0 Project Overview

Project Title: AI Recommendation Filter

Client: Nathan Henry

Team 33 Members: Kean Scott, Sia Dubey, Tanish Khurana, Luvish Ramchurn, Mashudul Biswas, Keandria Ong

The objective of this project is to design and implement a Chrome browser extension that uses **allostatic regulation** to dynamically filter and restrict different types of content recommended to users on YouTube. The extension will adapt content exposure based on user preferences and historical engagement patterns, aiming to support healthier viewing habits.

The project will deliver a functional **Minimum Viable Product (MVP)** by the end of the semester, along with an open-source repository containing source code, setup instructions, and documentation for future development.

1.1 Project Value Distribution

Component	% of Project Value
Working MVP with Core Objectives Achieved	90%
Public Git Repository with the Source Code, Documentation, + Testing Framework	10%

2.0 Core Objectives

The core objectives are the key features of the MVP and represent 90% of the overall project value. The percentages listed alongside each feature indicate its proportion of the total project value, collectively adding up to 90%. The MVP will deliver the following functionality:

1. **Dynamic YouTube Page Interaction (30%)**
    - The extension should dynamically update and modify the YouTube interface in real time based on the user's active regulation settings.
2. **Content Classification (25%)**
    - Implement an LLM-based classifier to evaluate YouTube video metadata (titles, descriptions, tags).
    - Classify videos into predefined content categories (e.g., political, violent, sexual, educational).
    - Cache results + prioritise local storage when possible to improve user experience.
3. **User Control Panel/ Dashboard (25%)**
    - Key Features:
        - View the *allostatic regulator graph* showing regulation levels for each content category over time.
        - Allow users to control what content categories are being regulated.
        - Allow users to control the degree to which each type of content is regulated (e.g. using sliders)

        - Support user-friendly language and visuals to ensure accessibility for non-technical users.
    - Additional Features:
        - Allow users to add new content categories
        - Include additional analytics (e.g. days/times where regulation is strongest)
4. **Extension Popup (10%)**
    - Key Features:
        - Display a compact popup when the extension icon is clicked in the browser toolbar.
        - Include:
            - An on/off toggle to enable or disable the extension.
            - A button linking to the full dashboard/control panel.
    - Additional Features:
        - Show a small analytics visual (e.g., allostatic regulator trends over the last 6 hours).

3.0 User Interaction + Design

The dashboard/ control panel will be developed to align as closely as possible with the approved design created by the team and endorsed by the client. While adherence to the design is important, the implementation of the MVP's key features and functionality will take priority over visual styling if time constraints arise. The extension popup design will follow the same theme as the dashboard/ control panel.

Link to Figma Design

Full design shown in next page:


Hi, Username
Overview
12 +
Hours Saved This Week
8% from last week
25 %
Of Content Removed
6% from last week
82
Videos Blocked This Week
4
Categories Blocked
Political
Most Blocked Content
452
Videos Blocked In Total
Last X Hours
100
80
60
40
20
0
10am
11am
12am
01am
02am
03am
04am
05am
06am
Categories:
Total Days Blocked
1. Category 1 Name
32 Days
2. Category 2 Name
28 Days
3. Category 3 Name
14 Days
4. Category 4 Name
14 Days
Control Panel
Content Categories
Category 1
Category 1 Description
Block All Content
Yes
No
Filter Strength
Weak
Category 2
Category 2 Description
Block All Content
Yes
No
Filter Strength
Moderate
Category 3
Category 3 Description
Block All Content
Yes
No
Filter Strength
Strong
Category 4
Category 4 Description
Block All Content
Yes
No
Category 5
Category 5 Description
Category 6
Category 6 Description
+ Create New Category
Video Types
Short Form Content
Type 1 Description
Block All Content
Yes
No
Type 2
Type 2 Description

4.0 Technology Stack

The Chrome browser extension will be developed using a flexible, non-restrictive technology stack to allow for future expansion and integration of additional tools if required. The primary expected technologies include:

- **JavaScript** – Core extension functionality and interactive features.
- **HTML & CSS** – Structure and styling of the extension interface.
- **Tailwind CSS** – Utility-first CSS framework for rapid, consistent, and responsive UI development.
- **Python** – Backend scripting, data processing, and API integrations as needed.

This open approach ensures the project can adopt additional technologies or frameworks to meet evolving requirements during development.

5.0 Deliverables

Per the unit guidelines, the team will complete 3 sprints before the end of the semester. By the end of the **third sprint** (5:00pm Monday 13th October), the team will provide:

1. **A functional MVP (90% of Project Value)** of the Chrome browser extension with the features listed in 2.0 Core Objectives and the design depicted in 3.0 User Interaction + Design.
2. **An open-source Git repository (10% of Project Value)** with the following:
    - Source code
    - Setup and installation instructions
    - Testing framework to validate functionality and ensure maintainability after the end of the project, with testing instructions
    - User + Developer documentation
3. **A System Demonstration** of the project.

6.0 Timeframe & Constraints

- Following the unit guidelines, the Project Deliverables will be due at 5:00pm on Monday, 13th October 2025.
- Development will follow an **Agile methodology**, allowing scope adjustments where necessary.

- Given semester time constraints, any proposed scope changes will be reviewed for feasibility before implementation.
- Optional features will only be implemented if time permits after the MVP is completed.

7.0 Optional Features

The following features are considered desirable but are **not part of the MVP**. An implementation of any of the following will be worth 5% of the total project value. They may be implemented if resources allow:

1. Include a step-by-step user tutorial upon installation.
2. Regulation for short vs. long-form content (e.g., YouTube Shorts vs. standard videos).
3. Import/export settings as a JSON file for portability between devices.
4. Context sensitivity - dynamically adjust regulation during vulnerable times (e.g., work hours).
5. Theme toggle between light mode and dark mode.

8.0 Exclusions

The following are outside the scope for the MVP:

- Integration with platforms other than YouTube.
- Compatibility with browsers other than Google Chrome
- Mobile application development.
- Full-scale machine learning model training from scratch (pre-trained models will be adapted).

9.0 Success Criteria

The project will be considered successful if:

- Users can successfully install the extension and use it without requiring additional technical assistance.
- The extension accurately classifies (through the use of an LLM) and regulates YouTube webpage content according to user settings.
- The UI is clear, intuitive, and visually appealing.
- Documentation enables future contributors to extend or maintain the project.
- The project will be publicly available and open source.